Molishka Sahu

AI-Focused Full-Stack Developer | Python, TypeScript, React, FastAPI, Backend Systems & Automation

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PROFESSIONAL SUMMARY

B.Tech CSE (AI) undergraduate at NIET Greater Noida building full-stack web applications, RESTful backend services, automated data pipelines, and AI tooling using Python, TypeScript, React, Next.js, FastAPI, and PostgreSQL. Delivered deployed platforms across scholarship verification, semantic search ranking, support ticketing, and LLM execution tracing backed by automated tests.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL, C, HTML5, CSS3

Frameworks & Libraries: React, Next.js, FastAPI, Vite, Tailwind CSS, SQLModel, SQLAlchemy, Pydantic, Scrapy, BeautifulSoup, Streamlit, Bootstrap 5

AI Tooling & Retrieval: Gemini API (Google Search Grounding), Sentence Transformers, Scikit-learn, LLM Evaluation, Prompt Engineering, ReAct Agent Tracing

Databases & Data Pipelines: PostgreSQL (pgvector), SQLite, Alembic, Relational Database Design, Schema Validation, Automated Data Extraction

Developer Tools & Testing: Git, GitHub, Docker, Linux, Playwright, Pytest, Unittest, Vitest, REST APIs, VS Code

Soft Skills: Technical Documentation, Problem Solving, Critical Thinking, Decision Making, Leadership, Time Management

PROJECTS

Atlas Engine [Live Demo | GitHub] — **Nov 2026**

Python, FastAPI, Scrapy, React, Tailwind CSS, BeautifulSoup, SQLite

- Built a 7-stage automated scholarship discovery and extraction pipeline using Python, Scrapy, BeautifulSoup, FastAPI, and SQLite across 11+ institutional domains and 20 providers, persisting 24 records and 149 traceable field-level evidence snippets.
- Engineered a deterministic 0–100% confidence scoring engine (95%+ verified threshold) with TLD/domain allowlist validation, differential change tracking, React/Streamlit inspection UIs, and 10/10 passing unit tests (unittest).

EduConnect [Live Demo | GitHub] — **Nov 2026**

Next.js, TypeScript, Tailwind CSS, FastAPI, Python, SQLAlchemy, PostgreSQL, Sentence Transformers

- Built a full-stack K-12 education leadership discovery and explainable hybrid ranking platform using Next.js 15, TypeScript, FastAPI, SQLAlchemy/Alembic, and PostgreSQL (pgvector) / SQLite with JWT auth and CSV/XLSX/JSON exports.
- Implemented Sentence Transformers (BAAI/bge-small-en-v1.5) semantic cosine retrieval, robots.txt-compliant ingestion adapters, and evidence-confidence scoring verified by 16/16 backend pytest, 6/6 Vitest, and Playwright E2E tests.

TriageDesk [GitHub] — **Oct 2026 – Nov 2026**

Python, FastAPI, SQLModel, Pydantic, PostgreSQL, SQLite, PyJWT, Docker, Bootstrap 5, JavaScript

- Engineered a support ticketing and SLA management system backed by 8 relational SQLModel tables, Docker Compose (PostgreSQL 16 / SQLite), Bearer JWT and SHA-256 X-API-Key authentication, and a Bootstrap 5 workspace.
- Implemented 3-tier RBAC (CUSTOMER, AGENT, ADMIN), ticket state-transition validation, MTTA/MTTR SLA analytics, bulk triage, internal staff notes, and immutable audit logging verified by 5/5 passing pytest tests.

Influencer Outreach Engine [GitHub] — **Nov 2026**

React, Vite, JavaScript, CSS3, Gemini API, REST APIs

- Developed a live creator-research and outreach workspace using React 19, Vite, and the Gemini Interactions API with Google Search grounding to discover car-rental creators and generate outreach sequences via parallel async calls.
- Implemented LLM-evaluated fit scoring across 4 signals (relevance, engagement, accessibility, conversion potential) alongside a Vite dev/preview middleware proxy (/api/gemini) keeping API credentials out of the client bundle.

NeuroTrace [Live Demo] — **Nov 2026**

JavaScript, HTML5, CSS3, LLM Evaluation, ReAct Agent Tracing, REST APIs

- Deployed a browser-based ReAct execution trace visualizer and LLM-as-a-Judge evaluation workspace supporting Groq, OpenAI, and OpenRouter endpoints with multi-turn session memory and JSON trace export.
- Implemented a 5-stage reasoning, local tool-execution (calculator, get_time), and evaluation loop triggering automated reflection when judge scores fall below 75%, tracking latency, token usage, and guardrail errors.

EDUCATION

Noida Institute of Engineering and Technology (NIET) — **2025 – 2029 (Expected)**

Bachelor of Technology (B.Tech) in Computer Science and Engineering (Artificial Intelligence) | Greater Noida, Uttar Pradesh

City Montessori School (CMS), Lucknow

Class XII (Senior Secondary): 82.25% | Class X (Secondary): 87%